Differential Dynamic Programming

Nguyen Van Thanh

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1 Preliminaries

1.1 Problem

Consider the following discrete-time trajectory optimization problem:

$$\begin{aligned} \min_{\boldsymbol{X},\boldsymbol{U}} \quad & J(\boldsymbol{X},\boldsymbol{U}) = \phi_T(\mathbf{x}_T) + \sum_{k=0}^{N-1} L(\mathbf{x}_k,\mathbf{u}_k) \\ \text{subject to} \quad & \mathbf{x}_k = \boldsymbol{f}(\mathbf{x}_{k-1},\mathbf{u}_{k-1}), \quad k = 1,2,\ldots,N \end{aligned} \tag{1.1}$$

where $\mathbf{x}_k \in \mathbf{R}^n$, $\mathbf{u}_k \in \mathbf{R}^m$ are the state and control vectors at time step k. $\boldsymbol{X} = [\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_N]$ is the state trajectory, $\boldsymbol{U} = [\mathbf{u}_1,\mathbf{u}_2,\ldots,\mathbf{u}_{N-1}]$ is the control trajectory, $\phi_N : \mathbf{R}^n \to \mathbf{R}$ is the terminal cost function, $l : \mathbf{R}^n \times \mathbf{R}^m \to \mathbf{R}$ is the running cost function, and $\boldsymbol{f} : \mathbf{R}^n \times \mathbf{R}^m \to \mathbf{R}^n$ is the system dynamics function.

We denote the cost (return) function starting from step r as follows:

$$J_r = \phi_N(\mathbf{x}_N) + \sum_{k=r}^{N-1} l(\mathbf{x}_k,\mathbf{u}_k) \tag{1.2}$$

The **value function** is defined as the optimal value of the cost function starting from time step r:

$$V_r(\mathbf{x}_r) = \min_{\boldsymbol{X},\boldsymbol{U}} J_r \tag{1.3}$$

The problem 1.1 is now to minimize J_1 with respect to control sequence $\boldsymbol{U}$, given the initial state $\mathbf{x}_0$ and the system dynamics.

Example

For simplicity, consider the following linear, 1-dimensional system:

1.2 Second-order method

A norminal control sequence $\{\mathbf{u}_k, k = 1,2,\ldots,N-1\}$ is chosen and the corresponding state trajectory $\{\mathbf{x}_k, k = 1,2,\ldots,N\}$ is generated by forward simulating the system dynamics. These sequences are stored. Then we proceed, in reverse time, from $k = N-1$ to $k = 1$, to chose an optimal incremental control sequence:

$$\delta\mathbf{u}_k = \mathbf{g}_k(\delta\mathbf{x}_k) \tag{1.4}$$

valid for a small region about the initial trajectory and which improve the control and state sequences to be calculated. An incremental $\delta\mathbf{x}_k, \delta\mathbf{u}_k$ will lead to an incremental $\triangle V_k$ and we want to optimize it, e.g:

$$\begin{aligned} \delta\mathbf{u}_k &= \arg\min_{\delta\mathbf{u}_k} \triangle J_k(\delta\mathbf{x}_k, \delta\mathbf{u}_k) \\ \text{subject to:} \quad \delta\mathbf{u}_k &= \mathbf{g}_k(\delta\mathbf{x}_k) \end{aligned} \tag{1.5}$$

Since from 1.1, $\delta\mathbf{x}_{k+r}$ is a function of $\delta\mathbf{x}_k, \delta\mathbf{u}_k, \ldots, \delta\mathbf{u}_{k+r-1}$, then the incremental in the cost function starting from step k, $\triangle V_k$ is a function of $\delta\mathbf{x}_k, \delta\mathbf{u}_k, \ldots, \delta\mathbf{u}_{N-1}$ and if we select the incremental $\delta\mathbf{u}_k$ as in 1.4, then the optimal increment $\triangle V_k$ is a function of $\delta\mathbf{x}_k$:

$$\triangle V_k(\delta\mathbf{x}_k) = \min_{\delta\mathbf{u}_k,\ldots,\delta\mathbf{u}_{N-1}} \triangle J_k(\delta\mathbf{x}_k, \delta\mathbf{u}_k \ldots, \delta\mathbf{u}_{N-1}) \tag{1.6}$$

Of course $\triangle V_k(\delta\mathbf{x}_k)$ is a function of $\mathbf{x}_1$ and initial control sequence $\{\mathbf{u}_k, k = 1, 2, \ldots, N-1\}$. If we assume that $\delta\mathbf{x}_k$ is sufficiently small, then we can approximate $\triangle V_k^o(\delta\mathbf{x}_k)$ by a second-order Taylor expansion:

$$\triangle V_k(\delta\mathbf{x}_k) = \frac{1}{2}\delta\mathbf{x}_k^T J_{xxk} \delta\mathbf{x}_k + \delta\mathbf{x}_k^T J_{xk} + \mathbf{a}_k \tag{1.7}$$

where we write $J_{xxk} = \partial^2 J_k / \partial\mathbf{x}_k^2$, second derivative of $J_k(\mathbf{x}_k)$ for short.
Using the Dynamic Programming principle we have:

$$\triangle V_k(\delta\mathbf{x}_k) = \min_{\delta\mathbf{u}_k} \left[\triangle L(\delta\mathbf{x}_k, \delta\mathbf{u}_k) + \triangle V_{k+1}^o(\delta\mathbf{x}_{k+1})\right] \tag{1.8}$$

For simplicity, we consider the case when $\delta\mathbf{x}_k$ and $\delta\mathbf{u}_k$ are scalars, and expand $\triangle L(\delta\mathbf{x}_k, \delta\mathbf{u}_k)$ to the second order Taylor expansion:

$$\triangle L(\delta\mathbf{x}_k, \delta\mathbf{u}_k) = \frac{1}{2}\begin{bmatrix}\delta x_k \\ \delta u_k\end{bmatrix}^T \begin{bmatrix} L_{xx} & L_{xu} \\ L_{ux} & L_{uu}\end{bmatrix}_k \begin{bmatrix}\delta x_k \\ \delta u_k\end{bmatrix} + \begin{bmatrix} L_x \\ L_u\end{bmatrix}_k^T \begin{bmatrix}\delta x_k \\ \delta u_k\end{bmatrix} + a_k \tag{1.9}$$

Appendix

A KKT Conditions in Optimization Problems

A.1 The Lagrangian Function

Consider the following optimization problem:

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & g_i(\mathbf{x}) \leq 0, \quad i = 1, \ldots, m \\ & h_j(\mathbf{x}) = 0, \quad j = 1, \ldots, p \end{aligned} \tag{A.1}$$

where the feasible domain $\mathcal{D} = \mathbf{dom}(f) \cap \bigcap_{i=1}^{m} \mathbf{dom}(g_i) \cap \bigcap_{j=1}^{p} \mathbf{dom}(h_j)$ is non-empty, e.g $\mathcal{D} \neq \emptyset$.

We define the Lagrangian $L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) : \mathbf{R}^n \times \mathbf{R}^m \times \mathbf{R}^p \to \mathbf{R}$ as:

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^{p} \mu_j h_j(\mathbf{x}) \tag{A.2}$$

where $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_m)$ and $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_p)$ are the Lagrange multipliers associated with the inequality and equality constraints, respectively. In a matrix form, we can write:

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}) + \boldsymbol{\mu}^T \mathbf{h}(\mathbf{x}) \tag{A.3}$$

where $\mathbf{g}(\mathbf{x}) = (g_1(\mathbf{x}), \ldots, g_m(\mathbf{x}))^T$ and $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), \ldots, h_p(\mathbf{x}))^T$ are vectors of the inequality and equality constraint functions, respectively.

If $\lambda_i \geq 0$ for all $i = 1, \ldots, m$, then it is easy to show that:

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \leq f(\mathbf{x}) \quad \forall \mathbf{x} \in \mathcal{D}, \lambda_i \geq 0 \tag{A.4}$$

as $\forall x \in \mathcal{D}, \lambda_i \geq 0, g_i(\mathbf{x}) \leq 0 \implies \lambda_i g_i(\mathbf{x}) \leq 0$ and $\forall x \in \mathcal{D}, h_j(\mathbf{x}) = 0 \implies \mu_j h_j(\mathbf{x}) = 0$.

Therefore, we have:

$$f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}) + \boldsymbol{\mu}^T \mathbf{h}(\mathbf{x}) \leq f(\mathbf{x}) \quad \forall \mathbf{x} \in \mathcal{D} \tag{A.5}$$

Taking the infimum over $\mathbf{x} \in \mathcal{D}$ on both sides of A.4, we have:

$$\inf_{\mathbf{x} \in \mathcal{D}} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \leq \inf_{\mathbf{x} \in \mathcal{D}} f(\mathbf{x}) \tag{A.6}$$

Given $\boldsymbol{\lambda}, \boldsymbol{\mu}$, the formula $\inf_{\mathbf{x} \in \mathcal{D}} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ is called the Lagrange dual function, denoted as $g(\boldsymbol{\lambda}, \boldsymbol{\mu}) : \mathbf{R}^m \times \mathbf{R}^p \to \mathbf{R}$:

$$g(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x} \in \mathcal{D}} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \tag{A.7}$$

And again, if $\lambda_i \geq 0$ for all $i = 1, \ldots, m$, we have:

$$g(\boldsymbol{\lambda}, \boldsymbol{\mu}) \leq \inf_{\mathbf{x} \in \mathcal{D}} f(\mathbf{x}) \tag{A.8}$$